```
Mlaby 7048 AgSp1 glycine rm1b 2                PGTTPGTVTGPDGKP.TKFIVPTGAFTTPGSIPGPDGKPIR.VEPAGPGTTPGAETGPDGKIN......KLVVPTTPKEP.....   72
Mlaby 7048 AgSp1 glycine rm1a 2                PGTTPGTVTGPDGKP.SKYIVPLGAYTTPGSIPGPDGKPIH.VQPAGPGTTPGTITDSQGNVD......KIILPTTPK..GQQ..   73
Mlaby 7048 AgSp1 glycine rm1a 4                PGTTPGVVTGPDGQP.VQFIVPQGAFTTPGSIPGPNGKRIH.VKPAGPGTTPGAK................................   53
Mlaby 7048 AgSp1 glycine rm1a 13               PGTTPGVVTGPGGEP.LRFVVPKGAFTTPGSIPGPNGKPIH.VGPSGPGTTPGAKTDSDGNID......SIVLPSTPKDSGPGF.   76
Mlaby 7048 AgSp1 glycine rm2a 21               ......................................................................LVVPTTTP..G....    9
Mlaby 7048 AgSp1 glycine rm2a 22               PGSTPGTVTGPDGMP.K....................................................................   16
Mlaby 7048 AgSp1 glycine rm2b 29               .......................GAFTTPGSIPGPDGKPIH.VQPAG......................................   23
Msag 18528 AgSp1.2 glycine rm 1                PGTTPGTVTGPDGKP..R...................................................................   16
Msag 18528 AgSp1.2 glycine rm 16               .....................................................................................    0
Msag 18528 AgSp1.2 glycine rm 112              .....................................................................................    0
Msag 18528 AgSp1.2 glycine rm 113              PGTTPGTVTGPDGKP.KKFVVPKGAFTTPGSFSGPDGQPIH.VEPAGPGTTPGVQTGPDGKIN......KVVVPTTT..KGPQGP   75
Msag 18528 AgSp1.2 glycine rm 114              PGTTPGTVTGPDGKP.KKFVVPKGAFTTPGSFSGPDGQPIH.VEPAGPGTTPGVQTGPDGKIN......KVVVPTTTK.......   70
Msag 18528 AgSp1.2 glycine rm 115              ...........SDGKP.QKFVIPKGAFTTPGFFPGPDGQPIH.VEPAK......................................   35
Varen 6869 AgSp1.2 glycine rm1 1               PGTTPGTITRHDGKP.KEFIIPIGAFTTPGSFPGPDGRPIY.VEPAGPGTTPGAITGPDGKIY......KIIVPTTQK.......   70
Varen 6869 AgSp1.1 glycine rm1 1               .................KFIIPNGAFRTPGSVPGPDGEPIS.VEPAGPGTTPGTETGPDGKIR......KVVIPTTPK..GPH..   57
Varen 6869 AgSp1.2 glycine rm1 2               PGTTPGTITGSDGKP.TKFIIPMGAFTTPGSMPGPDGRPIR.VEPAGPGTTPGVVTRHDGKIY......KIIVPAMQMGSDE...   74
Varen 6869 AgSp1.1 glycine rm1 2               PGTTPGTVTGRDGKP.RKFIVPNGAFTTPGSFPGPDGEPIS.VEPAGPGTTPGIETDSDGKIK......KVVVPSTPK..GPEGP   75
Varen 6869 AgSp1.2 glycine rm1 3               PGTTPGTVTGK..........................................................................   11
Varen 6869 AgSp1.2 glycine rm1 10              ......................................................................IIVPTI.........    6
Varen 6869 AgSp1.1 glycine rm1 11              .....................................................................................    0
Varen 6869 AgSp1.2 glycine rm1 11              PGTTPGTVTGKDGKP.TKFIIPIGAFTTPGSIPGPDGRPIR.VEPAGPGTTPGVVTRPDGKIY......KIIVPTIHMGSDE...   74
Varen 6869 AgSp1.2 glycine rm1 12              PGTTPGTVTGNDGKP.TKFIIPIGAFTTPGSIPGPDGRPIR.VEPAGPGTTPGVVTRPDGKIY......KIIVPTIHMGSDE...   74
Varen 6869 AgSp1.2 glycine rm1 13              PGTTPGTVTGKDGKP.TKFIIPIGAFTTPGSIPGPDGRPIR.VEPAGPGTTPGVVTRPDGKIY......KIIVPTIHMGSDE...   74
Varen 6869 AgSp1.2 glycine rm1 14              PGTTPGTVTGKDGKP.IKFIIPIGAFTTPGSIPGPDGRPIR.VEPAG......................................   45
Aarg AgSp1 FC511 glycine rm1 1                 PGTTPGTITGPDGKP.TKFIIPLGAFTTPGSIPGPDGRPIH.VQPAGPGTTPGAETGADGKIN......KIIVPTTPKS......   71
Aarg AgSp1 FC511 glycine rm1 2                 PGTTPGTVTGPDGKP.KKFVVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTGPDGKIN......KLVVPTTTTPKGPVGP   77
Aarg AgSp1 FC511 glycine rm1 3                 PGTTPGTVTGSDGKP.KKFIVPK..............................................................   22
Aarg AgSp1 FC511 glycine rm1 13                .......................GAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTGPDGKIN......KIVVPTTTTSKGPVGP   55
Aarg AgSp1 FC511 glycine rm1 15                PGTTPGTVTGPDGKP.KKFVVPQGAFTTPGSIPGPDGNPIP.VEPAGPGTTPGTQTGPDGKIN......K...............   62
Aarg AgSp1 FC511 glycine rm2 6                 PGTTPGTITGPDGNP.TKFIVPLGAFTTPGSIPGPDGKPIH.VQPAGPGTTPGAETGPDGKIN......KIILPTTPKRPSYP..   75
Aarg AgSp1 FC511 glycine rm2 7                 PGTTPGTVTGPDGQP.VKFIVPQGAFVTPGTILGPDGKPIP.VEPQG......................................   45
Ncruc 58532and13532 AgSp1 glycine rm 1        PGTTPGTVTGPDGKP.TKFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP..KGPGG.   74
Ncruc 58532and13532 AgSp1 glycine rm 2        PGTTPGTVTGPDGKP.TKFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP..KGPGG.   74
Ncruc 58532and13532 AgSp1 glycine rm 3        PGTTPGTVTGPDGKP.TKFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP..KGPGG.   74
Ncruc 58532and13532 AgSp1 glycine rm 4        PGTTPGTVTGPDGKP.TKFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP..KGPGG.   74
Ncruc 58532and13532 AgSp1 glycine rm 5        PGTTPGTVTGPDGKP.TKFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTP..KGPGG.   74
Ncruc 58532and13532 AgSp1 glycine rm 6        PGTTPGTVTGPDGKP.TKFVVPLGAFTTPGSIPGPDGQPIH.VEPAGPGTTPGVQTGPDGK.......................   59
Lcorn 18104 AgSp1.1 glycine rm1 1              PGTTPGTVTGPDGKP.IKFIVPTGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTGPDGKIN......KLVLPTT..PKAPQGP   75
Lcorn 18104 AgSp1.1 glycine rm1 37             .GTTPGTVTGPDGKP.TKFIVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAETGPDGKIN......KLVVPTTPK..GPQGP   74
Lcorn 18104 AgSp1.1 glycine rm1 38             PGTTPGTVTGPDGKP.NKFIVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTGPDGKIN......KLVVPTT..PKGPQGS   75
Lcorn 18104 AgSp1.1 glycine rm1 39             PGTTPGTVTGPDGKP.TKFIVPKGAFTTPGSIPGPDGKPIH.VEPAG......................................   45
Amarm 50219 AgSp1.1 glycine rm1 1              PGTTPGTITGPDGKP.TKFIVPYGAFTTPGSIPGPDGRPIH.VEPAGPGTTPGAETGPDGKVN......KIYLPTTPK..GPGGP   75
Amarm 50219 AgSp1.1 glycine rm1 2              PGTTPGTVTGPDGKP.TKFIVPTGAFTTPGSIPGPDGRPIH.VEPAGPGTTPGAETGPDGKIN......KIYLPST..PKGPGGP   75
Amarm 50219 AgSp1.1 glycine rm1 3              .....................................................................................    0
Amarm 50219 AgSp1.1 glycine rm1 8              PGTTPGTVTGPDGKP.IKFIVPTGAFTTPGSIPGPDGRPIH.VEPAGPGTTPGAETGPDGKIN......KIYLPST..PKGPGGP   75
Amarm 50219 AgSp1.1 glycine rm2 1              PGTTPGTETGTDGKP.NKFVVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAETGPDGKIN......KIVVPTTTK..APTGP   75
Amarm 50219 AgSp1.1 glycine rm2 78             .....................................................................................    0
Amarm 50219 AgSp1.1 glycine rm2 89             .....................................................................................    0
Atri AgSp1 genomic glycine rm1 1               PGTTPGTITGPDGKP.TKFIIPLGAFTTPGSIPGPDGRPIH.VQPAGPGTTPGAETGPDGKIN......KIIVPTT.........   68
Atri AgSp1 genomic glycine rm1 2               PGTTPGTVTGPDGKP.KKFVLPKGAFTTPGSIPGPDGKPIH.VQPAGPGTTPGAQTGPDGKIN......KLVVPTTTTPKGPVGP   77
Atri AgSp1 genomic glycine rm1 47              .....................................................................................    0
Atri AgSp1 genomic glycine rm2 40              .................KFVLPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTGPDGKIN......KLVVPTTT.TP.....   55
Mhut 1271 AgSp1 glycine rm1 1                  PGTTPGTITGPDGKP.IQFIVPYGAFSTPGSIPGPDGIPIL.VEPAESGKTPGIETGPDGKIY......KIYLPIIQKGPDKHRT   77
Mhut 1271 AgSp1 glycine rm1 2                  PGTTPGTITGPDGRP.TKFIVPYGAFTTPGTIPGPDGKPIQ.VEPAGPGTTPGVERGDDGKVY......KIYLPLMHISPGELI.   76
Mhut 1271 AgSp1 glycine rm1 3                  PGTTPGTITGPDGKP.TKFIVPYGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGVKRGPDGKVY......KIYLPTTPK..GPGGP   75
Mhut 1271 AgSp1 glycine rm1 4                  PGTTPGTITGSDGRP.IQFIIPYGAYSTPGSIPGPDGTPIH.VEPAGPGTTPGIETGPDGKVY......KIYLPTTPK..GPGGP   75
Mhut 1271 AgSp1 glycine rm1 5                  PGTTPGTITGSDGRP.IKFIIPYGAYSTPGSIPGPDGTPIH.VEPAGPGTTPGIETGPDGKVY......KIYLPTRPK.....RP   72
Mhut 1271 AgSp1 glycine rm1 6                  PGTTPGTITGSDGRP.IKFIIPYGAYSTPGSIPGPDGTPIH.VEPAGPGTTPGVKTGPDGKVY......KIYLPTTPK..GPGGP   75
Mhut 1271 AgSp1 glycine rm1 7                  PGTTPGTLTGSDGRP.IKFIIPYGAFSTPGSIPGPDGTPIH.VEPAGPGTTPGVKTGPDGKVY......KIYLPTTPK..GPGGP   75
Mhut 1271 AgSp1 glycine rm1 8                  PGSTPGTITGPDGRP.TQFIIPFGAFTTPGSIPGPDGIPIH.VEPAG......................................   45
Mhut 1271 AgSp1 glycine rm3 17                 PGTTPGYVTDPDGQA.VKFIVPFDAFITPGYIPGPNGKPVR.VGPAG......................................   45
```

Mlaby 7048 AgSp1 glycine rm1b 2	..........TGPGGQPLNPQ..GPISPFGPGGQSM....NPLGPGGS..GG....QTTTPEPIPGPNGKPLQIVPAG	128
Mlaby 7048 AgSp1 glycine rm1a 2	.F.......................Y..............FP........VT....PKPTPTPVTGPGGKPIQIIPAG	101
Mlaby 7048 AgSp1 glycine rm1a 4	....	53
Mlaby 7048 AgSp1 glycine rm1a 13	.......QFPTGPGGQPLNPT..G..	91
Mlaby 7048 AgSp1 glycine rm2a 21	.........PTGPGGQPLNP........LGPRGQPL..........NPLGPGSPGGQTTTPAPVPGPDGKPLQIVPAG	60
Mlaby 7048 AgSp1 glycine rm2a 22	....	16
Mlaby 7048 AgSp1 glycine rm2b 29	....	23
Msag 18528 AgSp1.2 glycine rm 1	....	16
Msag 18528 AgSp1.2 glycine rm 16	.................................GPQGPGGYPLGPGGQ........T.....................	16
Msag 18528 AgSp1.2 glycine rm 112	................................RGPQGPGGYPLGPGSPFGPKGPGGQTTTPASVQGPDGEPIQVEPAG	46
Msag 18528 AgSp1.2 glycine rm 113	GG......................................YPLGPGSPFGPKGPGGQTTTPASVQGPDGEPIQVEPAG	115
Msag 18528 AgSp1.2 glycine rm 114	....	70
Msag 18528 AgSp1.2 glycine rm 115	....	35
Varen 6869 AgSp1.2 glycine rm1 1	....PETHPAGRPIQIVPAG	86
Varen 6869 AgSp1.1 glycine rm1 1	....GPGKMFGPMRPRRQRTTPPSVEGPNGKPIQIEPAG	92
Varen 6869 AgSp1.2 glycine rm1 2	....QGM....NFATTTPIPGPGGRPIQIEPAG	99
Varen 6869 AgSp1.1 glycine rm1 2	GGY..PMRPR..	83
Varen 6869 AgSp1.2 glycine rm1 3	....	11
Varen 6869 AgSp1.2 glycine rm1 10	....HMGSDEQGMNFVTTTPIPGPGGRPIQIEPAG	37
Varen 6869 AgSp1.1 glycine rm1 11	...........GPEG......................PGGYPMGPGNPFRPMGPGRQRTTPSSVRGPKGKPIQIEPAG	45
Varen 6869 AgSp1.2 glycine rm1 11	....QGM....NFATTTPIPGPGGRPIQIEPAG	99
Varen 6869 AgSp1.2 glycine rm1 12	....QGM....NFATTTPIPGPGGRPIQIEPAG	99
Varen 6869 AgSp1.2 glycine rm1 13	....QGI....NFATTTPIPGPGGRPIQIEPAG	99
Varen 6869 AgSp1.2 glycine rm1 14	....	45
Aarg AgSp1 FC511 glycine rm1 1	...........................PEEPSGRPM....YPFGPGSPYGPG....EQTTTTPIPGPDGKPLQIEPAG	114
Aarg AgSp1 FC511 glycine rm1 2	GG................MPL..SPYYPQGPGGQPM....NPFAPGSPYGPG....EQTTTTPIPGPDGKPLQIEPAG	129
Aarg AgSp1 FC511 glycine rm1 3	....	22
Aarg AgSp1 FC511 glycine rm1 13	DG.............QPMYPS..GP.......................Q...GPG.GQTTTTPIPGPDGKPIQIEPAG	91
Aarg AgSp1 FC511 glycine rm1 15	....	62
Aarg AgSp1 FC511 glycine rm2 6	....IPTTTTPIPGP.GQPIQIIPAG	96
Aarg AgSp1 FC511 glycine rm2 7	....	45
Ncruc 58532and13532 AgSp1 glycine rm 1	....PGFNF.....GTPATTPTPIQGPGGKPIQVVPAG	103
Ncruc 58532and13532 AgSp1 glycine rm 2	....PGFNF.....GTPATTPTPIPGPGGKPIQVVPAG	103
Ncruc 58532and13532 AgSp1 glycine rm 3	....PGFNF.....GTPATTPTPIPGPGGKPIQVVPAG	103
Ncruc 58532and13532 AgSp1 glycine rm 4	....PGFNF.....GTPATTPTPIPGPGGKPIQVVPAG	103
Ncruc 58532and13532 AgSp1 glycine rm 5	....PGFNF.....GTPATTPTPIPGPGGKPIQVVPAG	103
Ncruc 58532and13532 AgSp1 glycine rm 6	....	59
Lcorn 18104 AgSp1.1 glycine rm1 1	GI.....................NPYYPMRPGR...	87
Lcorn 18104 AgSp1.1 glycine rm1 37	GL.....................NPYYPMGPEGKPM....YPYGPGSPFGPQGPGGQT.TKTPVPGPDGKPIQIVPAG	126
Lcorn 18104 AgSp1.1 glycine rm1 38	GQ..............................PL......FPFGPQ...GPV....GQTTTTPIPGPDGKPIQIVPAG	110
Lcorn 18104 AgSp1.1 glycine rm1 39	....	45
Amarm 50219 AgSp1.1 glycine rm1 1	GG...QTTPTPVPGPRGKPIQILPAG	98
Amarm 50219 AgSp1.1 glycine rm1 2	GG...QTTPTPVPGPR..........	88
Amarm 50219 AgSp1.1 glycine rm1 3	....GKPIQILPAG	10
Amarm 50219 AgSp1.1 glycine rm1 8	GF................NFA..TP................................ATTPTPIPGPGGKPIQVIPAG	103
Amarm 50219 AgSp1.1 glycine rm2 1	GG.............FPLGP....AGFPLGPGGQPI....YPYGPGSPYGPQGPGGKT.TTTPIPGPDGEPLSIVPAG	131
Amarm 50219 AgSp1.1 glycine rm2 78	........APTGPGGFPLRT........TGPGGQPI....YPYGPGSPYGPQGPGGK.....................	37
Amarm 50219 AgSp1.1 glycine rm2 89	........APTGPGGFPL..........PGPGGQPI....YPYGPGSPYGPQGPGGK.....................	35
Atri AgSp1 genomic glycine rm1 1	........................PKTPQGSDGQPM....YPFGPGSPYGPG....EQTTTTPIPGPDGKPLPIEPAG	114
Atri AgSp1 genomic glycine rm1 2	GG................MPL..SPYSPQGPGGQPM....YPFGPGSPYGPG....EQTTTTPIP.............	116
Atri AgSp1 genomic glycine rm1 47	....GPDGKPLQIEPAG	13
Atri AgSp1 genomic glycine rm2 40	.........................KGPLGPGGQPM....YPSGPQGTG.......GQTTTTPIPGPDGNPIQIEPAG	97
Mhut 1271 AgSp1 glycine rm1 1	K..RPSY.THVKDKPIQVIPAG	96
Mhut 1271 AgSp1 glycine rm1 2	....LGKASHGRPIQIIPAG	92
Mhut 1271 AgSp1 glycine rm1 3	GFYFG..MPQYIPGPGGRPIQIVPAG	99
Mhut 1271 AgSp1 glycine rm1 4	GFYFG..MPQYIPGPGGRPIQIVPAG	99
Mhut 1271 AgSp1 glycine rm1 5	GFYFG..MPQYIPGPGGKPIQIEPAG	96
Mhut 1271 AgSp1 glycine rm1 6	GFYFG..MPQYIPGPGGKPIQIEPAG	99
Mhut 1271 AgSp1 glycine rm1 7	GFYFG..MPQNIPGPDGKPIQIVPAG	99
Mhut 1271 AgSp1 glycine rm1 8	....	45
Mhut 1271 AgSp1 glycine rm3 17	....	45

X non-conserved
X similar
X $\geq 50\%$ conserved